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SFRNet: Ultra-high-definition moiré pattern detection

Pengyue Xu [a,b,1], Shuang Chen [a,b,1], Qixian Zhang [c,1], Ning Xie [a,b], Hongbin Wang [a,b,*]

[a] Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, 650500, Yunnan, China
[b] Key Laboratory of Artificial Intelligence in Yunnan Province, Kunming University of Science and Technology, Kunming, 650500, Yunnan, China
[c] School of Computer Science and Technology, Tongji University, Shanghai, 201804, China

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ABSTRACT

With the rapid spread of 4K displays and ultra-high-resolution imaging sensors, detecting moiré patterns in Ultra-High-Definition (UHD) images has become an important task for image restoration and distinguishing real scenes from screen-based spoofing attacks. However, existing research mainly focuses on low-resolution or synthetic moiré images. There is a lack of real-world UHD datasets, and current methods struggle to handle the drastic frequency variations and spatially non-uniform distribution of moiré patterns in 4K scenes. To fill this data gap, we built the Ultra-High-Definition Moiré Detection (UHDMD) dataset, which is the first real-world UHD dataset dedicated to 4K moiré detection. To address the difficulty in accurately extracting features caused by the complex texture properties of UHD moiré patterns, we propose the Scale-aware Fission Reorganization Network (SFRNet). This network mainly consists of two modules. One is the Scale-Aware Fusion Module (SAFM), which accurately captures multi-scale moiré features via cross-scale dynamic fusion. The other is the Fission Reorganization Module (FRM), which innovatively employs a patch-level Mixture of Experts (MoE) mechanism. By using a gating network to dynamically assign image blocks to specific expert branches, it extracts diverse local moiré features with optimal parameters, effectively overcoming the limitations of existing segmentation methods in identifying subtle local textures. We evaluated our model on the self-built UHDMD and the public MoireScape datasets, establishing a performance benchmark for the UHD moiré pattern detection task. Experimental results prove the effectiveness of SFRNet, and its performance significantly outperforms existing classic and state-of-the-art methods. Code and datasets are available at https://github.com/XuPengyue30/SFRNet.

1. Introduction

With the rapid development of display technology and high-performance imaging sensors, Ultra-High-Definition (UHD) devices like 4K screens and high-resolution smartphones have become widely adopted in daily life (Shishikui & Sawahata, 2021). However, the spatial aliasing between the digital camera sensor array and the screen's pixel grid often generates complex moiré patterns during capture (Ning et al., 2022; Ri et al., 2012; Yang et al., 2023), which severely reduce image quality and obscure original texture details (Sun et al., 2018; Yang et al., 2023; Yang & Tsai, 2000). Consequently, detecting moiré patterns is crucial for aiding demoiréing and evaluating image restoration (Chen et al., 2025; Yuan et al., 2020). Concurrently, the broader field of UHD image restoration has also witnessed remarkable advancements, propelled by the introduction of specialized large-scale benchmarks and highly efficient architectures (Wang et al., 2026, 2024, 2026). Furthermore, it plays a vital role in security-sensitive scenarios, such as financial payments and identity verification, by distinguishing real-world scenes from screen-based spoofing attacks (Patel et al., 2015; Xu et al., 2023). Therefore, moiré pattern detection in UHD images has become a task of significant research importance.

Although significant progress has been made in existing image segmentation and artifact detection tasks, these methods are difficult to apply directly to UHD moiré detection. For instance, Hu et al. (2020) utilized direction-aware spatial context features for shadow detection, Nah et al. (2017) employed multi-scale deep feature fusion for motion blur identification, Xu et al. (2024) proposed a video mirror detection network to locate high-reflection areas in dynamic scenes, and Zhao et al. (2024) used diffusion models to capture geometric projections between objects and lighting. These methods typically target tangible objects with specific physical boundaries or continuous low-frequency textures, making it difficult for them to capture the extremely high-frequency and spatially non-uniform moiré features in UHD images. However, research specifically dedicated to moiré pattern detection remains limited. Existing research are based on synthetic datasets, such as MoireScape (Yang et al., 2023), rely on low-resolution images or

* Corresponding author .
E-mail addresses: xupengyue30@gmail.com (P. Xu), whbin2013@kust.edu.cn (H. Wang).
[1] These authors contributed equally to this work.

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synthetic data generated via mathematical superposition. As shown in Fig. 1, synthetic data fail to provide the authentic features of real-world UHD moiré patterns (He et al., 2020). Furthermore, the distribution of moiré patterns in images is highly non-uniform. For instance, a local region may present high-frequency fine stripes, while an adjacent area may appear as sparse color casts (Liu et al., 2020; Yang et al., 2017). Such drastic variations in color, density, and frequency make it difficult for traditional segmentation models to handle these changing texture features (Hsiao & Sawchuk, 1989). This often leads to blurred detection edges or missed detections (Marmanis et al., 2018). Therefore, the moiré pattern detection task still faces severe challenges.

To address the lack of high-quality real-world benchmarks in existing research, we constructed the first Ultra-High-Definition Moiré Detection (UHDMD) dataset. Since synthetic data fail to simulate the nonlinear signal attenuation and optical interference inherent in real physical imaging (Brooks et al., 2019), we employed three distinct high-resolution smartphones to capture a total of 4625 image pairs from three types of digital displays under complex shooting environments. These images contain real-world lighting interference and sensor nonlinear distortions. As the first benchmark dedicated to 4K real-world moiré detection, UHDMD provides reliable data support for future algorithmic research.

Based on this dataset, to address the difficulty in accurately extracting the complex texture features of UHD moiré patterns, we propose the SFRNet network model. This network model precisely captures diverse local texture features of various sizes in UHD moiré patterns. Specifically, we designed the Scale-Aware Fusion Module (SAFM) and the Fission Reorganization Module (FRM). SAFM captures multi-scale moiré features via cross-scale dynamic fusion. Meanwhile, FRM incorporates a patch-level Mixture of Experts (MoE) mechanism (Masoudnia & Ebrahimpour, 2014), utilizing a routing network to dynamically assign partitioned image blocks to specific expert branches to extract diverse local patterns, effectively addressing the limitations of existing image segmentation methods in identifying subtle local textures. Based on the combination of cross-scale feature modulation and refined local processing, the proposed network model achieves excellent performance in complex moiré pattern detection tasks.

The contributions of this study are summarized as follows:

- **Dataset and Benchmark:** This study pioneers the exploration of moiré pattern detection in UHD images. We constructed the first UHDMD dataset, a real-world dataset for 4K scenes, filling the gap in non-synthetic UHD data resources and establishing a performance benchmark for the UHD moiré pattern detection task.
- **Innovative Methodology:** We propose SFRNet, an innovative network model framework specialized for UHD moiré detection. This network model effectively solves the detection challenges caused by the spatially non-uniform distribution of moiré patterns by organically combining SAFM for capturing multi-scale features and FRM based on a patch-level MoE architecture.
- **Effectiveness and Generalizability:** Experimental results on the self-built UHDMD dataset and the public MoireScape dataset demonstrate that SFRNet significantly outperforms state-of-the-art methods in detection accuracy, proving its superior effectiveness and favorable generalizability in diverse real-world scenarios.

2. Related work

In this section, we review the literature closely related to our study from three perspectives. First, we introduce the evolution of moiré pattern detection methods and analyze their current bottlenecks. Second, we discuss generic semantic segmentation architectures, which serve as foundational baselines for pixel-level localization, while highlighting their limitations in handling complex frequency anomalies. Finally, we review existing moiré image benchmarks to demonstrate the pressing need for constructing a real-world, ultra-high-definition dataset.

2.1. Moire pattern detection

Moiré pattern detection aims to locate artifact regions caused by frequency aliasing. This is fundamentally different from the traditional demoiréing task. Early research on moiré processing mainly focused on image restoration and denoising through frequency domain filtering or multi-scale Convolutional Neural Networks (CNNs) (He et al., 2020; Sun et al., 2018; Yu et al., 2022a). However, with the increasing demand for multimedia forensics, the task of pure detection has become more important. Moiré patterns serve as a key clue for distinguishing real natural images from images recaptured from screens (Yue et al., 2020). Therefore, they are widely applied in Face Anti-spoofing and anti-counterfeiting forensics (Garcia & de Queiroz, 2015; Peng et al., 2017).

Regarding detection methods, early works relied on hand-crafted frequency features such as FFT peak detection (Gonzalez & Woods, 2018; Jiang & Xu, 2001) to identify periodic patterns (Amidror, 2009). Recently, Yang et al. (2023) pioneered pixel-level moiré detection using deep multi-scale feature fusion. Although effective, such joint detection and demoiréing architectures strictly require perfectly paired clean images for training, which are physically difficult to obtain in real-world 4K screen-recaptured scenarios. Moreover, their complex dense unfolding mechanisms consume massive GPU memory, making them difficult to train and deploy for ultra-high-definition images. Therefore, a dedicated model focusing purely on real-world moiré segmentation with practical memory efficiency is highly needed. Meanwhile, other approaches often directly borrow strategies from translucent object detection tasks like shadow or rain removal (Hnewa & Radha, 2020). Unlike these artifacts, moiré patterns exhibit highly complex textures and vary significantly in size (He et al., 2021). General feature extractors often fail to distinguish high-frequency object details from moiré artifacts, leading to frequent false detections in texture-rich Ultra-High-Definition (UHD) images. To address this gap, our work abandons these generic artifact removal strategies and specifically targets the complex frequency properties of real-world UHD moiré patterns.

2.2. Generic semantic segmentation architectures

Since moiré pattern detection can be formally viewed as a pixel-level binary classification problem (moiré vs. background), general semantic segmentation models can be adapted for this task. Representative methods, such as the DeepLab series (Chen et al., 2018) and PSPNet (Zhao et al., 2017), focus on enlarging the receptive field to capture global context information. DeepLabV3+ introduces the Atrous Spatial Pyramid Pooling (ASPP) module to detect multi-scale features using atrous convolutions with different dilation rates, while PSPNet utilizes a pyramid pooling module to aggregate context from different regions. However, these models typically rely on deep backbone networks and large downsampling rates to extract semantic information. Although this downsampling strategy is beneficial for identifying semantic objects with fixed shapes (such as cars and pedestrians), it causes an irreversible loss of high-frequency detail signals in moiré detection (Xie et al., 2021; Xu et al., 2020), making it difficult for the models to locate subtle stripe boundaries.

To alleviate the loss of spatial information, SegNet (Badrinarayanan et al., 2017) and U-Net (Ronneberger et al., 2015) adopt a symmetric encoder-decoder structure. They use skip connections to transfer shallow detail features to the decoder to assist in boundary recovery. HRNet (Wang et al., 2021) avoids resolution loss caused by downsampling by maintaining high-resolution branches in parallel throughout the process. Although these methods preserve spatial fidelity to some extent, they lack specific mechanisms to perceive local frequency. Furthermore, recent models such as InSPyReNet (Kim et al., 2022) and BiRefNet (Zheng et al., 2024), despite their success in highly accurate salient object segmentation, are fundamentally optimized for extracting common object shapes. When processing 4K UHD images, moiré

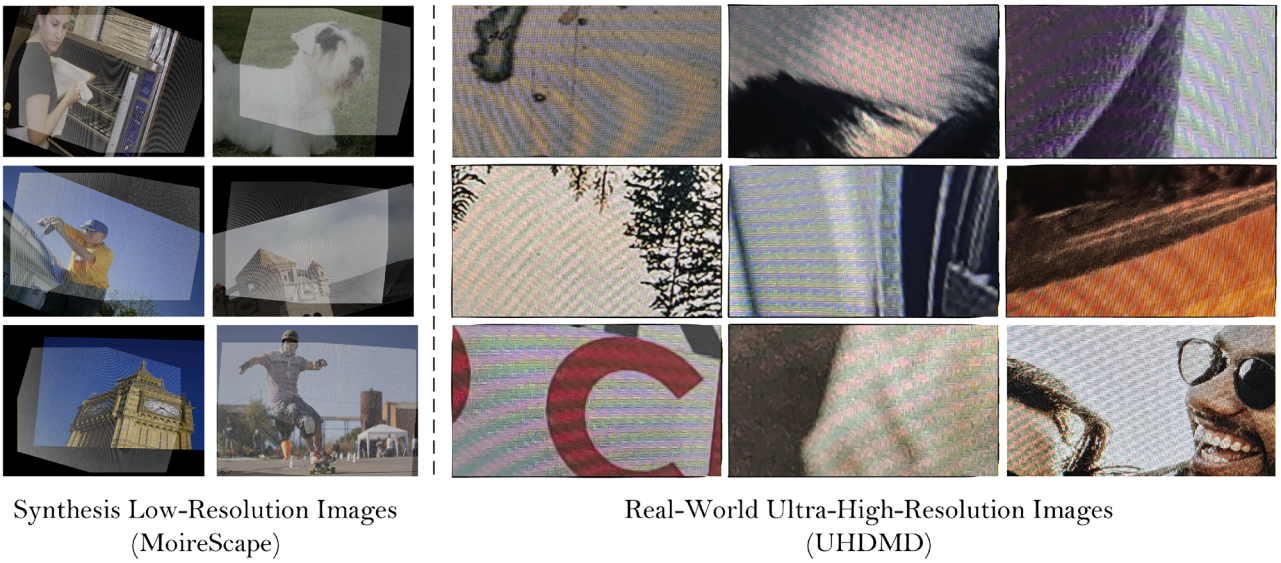

Fig. 1. Visual comparison between the synthetic MoireScape dataset and our proposed real-world UHDMD dataset. While MoireScape images are low-resolution and exhibit simple, uniformly distributed synthetic patterns, UHDMD captures authentic 4K screen-recaptured scenes featuring complex optical nonlinearities, diverse texture frequencies, and spatially non-uniform moiré artifacts.

patterns often exhibit features of various sizes and a spatially non-uniform distribution (Xiao et al., 2024). Consequently, these general and advanced models alike struggle to distinguish between semantic object edges and pure texture patterns within complex backgrounds, frequently leading to a large number of false detections or fragmented masks. Unlike these generic architectures, our proposed SFRNet introduces a patch-level Mixture of Experts (MoE) mechanism designed explicitly to perceive local frequency anomalies and handle the spatially non-uniform distribution of UHD textures.

2.3. Moire image benchmarks

High-quality datasets are the cornerstone of research. However, existing benchmarks exhibit limitations in authenticity, resolution, or task orientation. Therefore, they cannot fully meet the demands of Ultra-High-Definition (UHD) moiré detection (Liu et al., 2025). Specifically, synthetic datasets like LCDMoire (Yuan et al., 2019) and the detection-oriented MoireScape (Yang et al., 2023) rely on mathematical models. They simply overlay moiré patterns onto clean images. Although MoireScape addresses the detection task, it fails to simulate complex optical nonlinearities and lens distortions. This deficiency leads to severe domain shifts. Consequently, models struggle to generalize in complex real-world backgrounds.

Conversely, existing real-world benchmarks focus on the demoiréing task and suffer from low resolution. For instance, TIP2018 is restricted to 256×256 pixels (Sun et al., 2018). This limitation causes the loss of high-frequency features essential for detection. Similarly, while FHDMi achieves 1080p (He et al., 2020) and the recent UHDM dataset (Yu et al., 2022b) provides 4K images, they are inherently designed for image restoration rather than precise pattern localization, lacking the pixel-level masks required for quantitative detection evaluation. In summary, a real-world, ultra-high-resolution benchmark dedicated to detection is currently lacking. This gap severely hinders the development of robust algorithms for real-world 4K applications. Our constructed UHDMD dataset directly fills this gap, serving as the first real-world 4K benchmark tailored specifically for moiré pattern detection, thereby enabling the evaluation and advancement of specialized algorithms in this domain.

3. Method

3.1. Overall architecture

To address the challenging task of localizing complex and spatially non-uniform moiré artifacts in ultra-high-definition (UHD) images, this paper proposes a novel architecture named SFRNet. As illustrated in Fig. 2, the overall model presents an inverted pyramid structure. The input image is simultaneously fed into two functionally complementary pathways, f_1 and f_2. f_1 is used for global multi-scale feature extraction, and f_2 is used for local fine-grained perception. Specifically, the f_1 pathway adopts a symmetric encoder-decoder structure based on residual connections to capture global semantic information and multi-scale features. Skip connections are employed between the encoder and decoder, and each level embeds a Scale-Aware Fusion Module (SAFM) designed to capture moiré patterns of different sizes. Concurrently, the f_2 pathway consists of a specifically designed Fission Reorganization Module (FRM), which is explicitly dedicated to capturing local features and fine-grained frequency anomalies.

3.2. Scale-aware fusion module

Moiré patterns in Ultra-High-Definition (UHD) images exhibit features of various sizes and frequencies (Ri et al., 2012). Consequently, simultaneously capturing fine stripe textures and large color ripples is highly challenging. To address this, we propose the Scale-Aware Fusion Module (SAFM). As illustrated in Fig. 3, this module utilizes a spatial pyramid structure to extract multi-scale semantic features and dynamically modulates the feature weights at different scales. This mechanism ensures the precise recognition of complex moiré textures.

3.2.1. Multi-scale feature branching

To capture moiré features at different spatial resolutions, we construct a three-branch pyramid feature extraction structure. For the input feature $F_{in}^{SAFM} \in \mathbb{R}^{C\times H\times W}$, we first apply bilinear interpolation to generate a set of multi-scale inputs $X_k = D_k(F_{in}^{SAFM})$, where $D_k(\cdot)$ denotes the downsampling operation with a factor of $k \in \{1, 2, 4\}$ (with $k = 1$ being the original scale). These hierarchical features are first fed into a feature extraction unit $\mathcal{H}(\cdot)$. The extracted features are then spatially

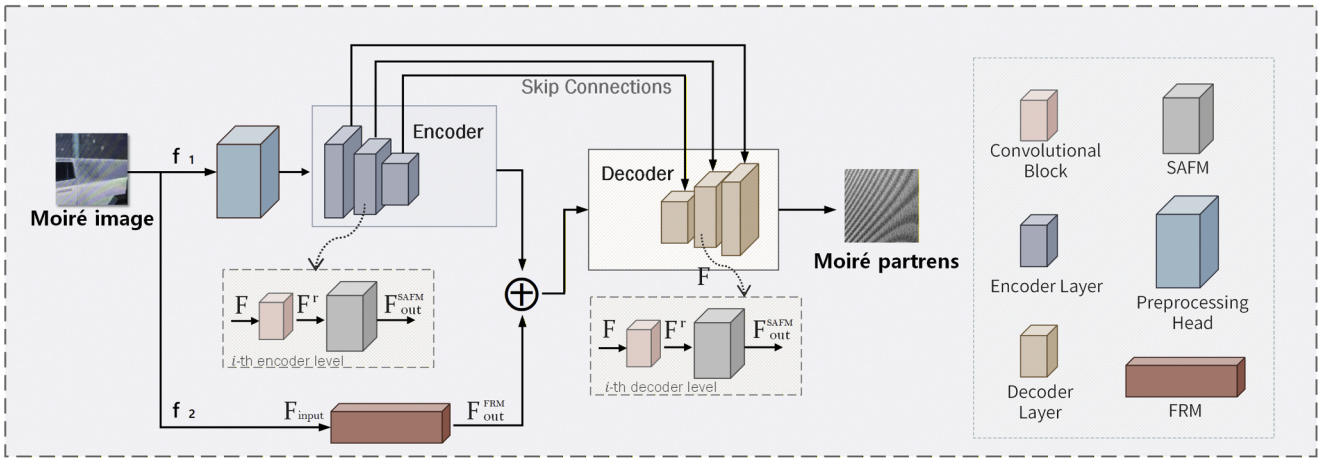

Fig. 2. Overall architecture of the proposed scale-aware fission reorganization network (SFRNet). The input image is processed through two complementary pathways: (f_1) utilizing a symmetric encoder-decoder structure with scale-aware fusion modules (SAFM) to capture global semantic and multi-scale features, and (f_2) featuring a fission reorganization module (FRM) dedicated to extracting fine-grained local frequency anomalies. The features from both pathways are fused to generate the final precise moiré detection mask.

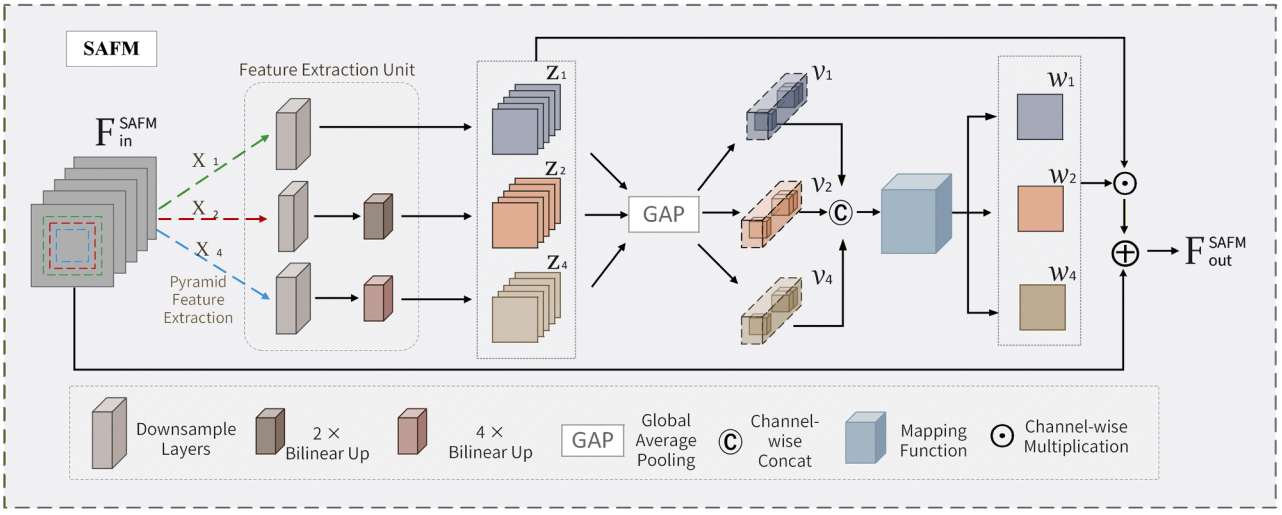

Fig. 3. Structure of the scale-aware fusion module (SAFM). The workflow consists of three main steps. First, the input feature is processed through a spatial pyramid to extract and align multi-scale features. Second, a global average pooling (GAP) layer and a mapping function are used to calculate an adaptive attention weight for each scale branch. Finally, these weighted features are fused and added back to the original input via a residual connection. This process effectively captures moiré patterns of different sizes, providing scale-enhanced features for the subsequent decoding stages.

aligned as follows:

$$Z_k = \mathcal{U}_k\big(\mathcal{H}(X_k)\big), \quad k \in \{1, 2, 4\}, \tag{1}$$

where $\mathcal{H}(\cdot)$ is the feature extraction mapping based on Dilated Dense connections, which is designed to effectively enlarge the receptive field. $\mathcal{U}_k(\cdot)$ represents the corresponding upsampling operation via bilinear interpolation, which restores the features Z_2 and Z_4 to the same spatial dimension $H \times W$ as Z_1, thereby achieving spatial semantic alignment for subsequent pixel-level fusion.

3.2.2. Feature enhancement and scale-aware fusion

Moiré patterns at different scales have distinct dominant frequencies, and simply adding multi-scale features does not effectively highlight the salient targets. Therefore, we introduce a cross-scale dynamic fusion mechanism, where the importance of each branch is adaptively adjusted by learning an input-dependent weight vector. First, the aligned multi-scale features Z_1, Z_2, Z_4 are processed by Global Average Pooling (GAP). This operation compresses global spatial statistics into channel descriptors $v_k \in \mathbb{R}^C$, calculated as follows:

$$v_k = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} Z_k(i, j). \tag{2}$$

Subsequently, the descriptors from the three scales are concatenated along the channel dimension and fed into a nonlinear mapping function $M(\cdot)$ to model the cross-scale channel correlation. Specifically, the mapping process consists of three 1×1 convolutional layers and two ReLU activation functions, and the final attention weights are normalized through a Sigmoid function σ:

$$[\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_4] = \sigma\big(M([v_1, v_2, v_4])\big), \tag{3}$$

where $[\cdot]$ denotes the concatenation operation. The generated weight vector $\mathbf{w}_k$ reflects the relative importance of the kth scale branch. Finally, the output of the SAFM, F_{out}^{SAFM}, is obtained by fusing the weighted multi-scale features and integrating a global residual connection:

$$F_{out}^{SAFM} = F_{in}^{SAFM} + \sum_{k \in \{1,2,4\}} \mathbf{w}_k \odot Z_k, \tag{4}$$

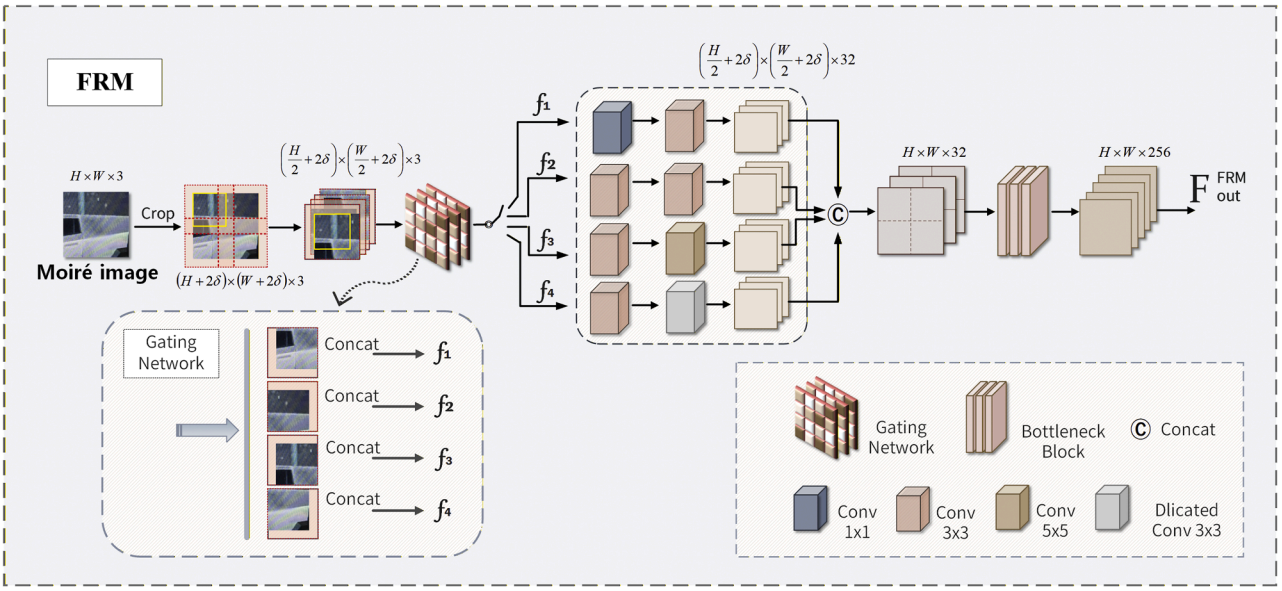

Fig. 4. Structure of the fission reorganization module (FRM). The workflow consists of three main steps. First, the input feature map is divided into overlapping sub-blocks to preserve local context. Second, a Gating Network routes each block to a specialized expert branch to extract diverse local moiré features. Finally, the processed blocks are cropped and stitched back to restore their original spatial dimensions. This reorganized feature is then fed into a bottleneck block for cross-channel fusion and integrated with the f_1 pathway for final mask prediction.

where $\odot$ denotes channel-wise multiplication broadcasted along the spatial dimensions. The addition of the original input F_{in}^{SAFM} acts as a residual pathway, which preserves the initial spatial details and facilitates stable gradient propagation. This design allows the network to adaptively enhance or suppress the feature responses of specific scales based on the sparsity or density of the moiré patterns, significantly improving the model's robustness in UHD detection tasks. The complete computational flow of this cross-scale dynamic fusion is detailed in Algorithm 1.

Algorithm 1 Cross-scale dynamic fusion in SAFM.

Input: Input feature map $F_{in}^{SAFM} \in \mathbb{R}^{C\times H\times W}$
Output: Scale-aware fused feature map $F_{out}^{SAFM} \in \mathbb{R}^{C\times H\times W}$

// Step 1: Multi-scale Feature Extraction and Alignment
for downsampling factor $k \in \{1,2,4\}$ **do**
  Downsample input: $X_k = D_k(F_{in}^{SAFM})$
  Extract hierarchical features: $H_k = \mathcal{H}(X_k)$
  Upsample to align spatial dims: $Z_k = \mathcal{U}_k(H_k)$ *// $Z_k \in \mathbb{R}^{C\times H\times W}$*
end for
// Step 2: Channel Descriptor Generation
for $k \in \{1,2,4\}$ **do**
  Compute global spatial statistics: $v_k = \mathrm{GAP}(Z_k)$ *// $v_k \in \mathbb{R}^{C\times 1\times 1}$*
end for
// Step 3: Cross-Scale Attention Weight Calculation
Concatenate descriptors: $V = \mathrm{Concat}(v_1, v_2, v_4, \mathrm{dim} = C)$
Non-linear mapping via three 1×1 convolutions:
  $V' = \mathrm{ReLU}(\mathrm{Conv}_{1\times 1}(V))$
  $V'' = \mathrm{ReLU}(\mathrm{Conv}_{1\times 1}(V'))$
  $W = \sigma(\mathrm{Conv}_{1\times 1}(V''))$ *// σ is Sigmoid function*
Split into scale-specific weights:
  $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_4 = \mathrm{Split}(W, \mathrm{chunks} = 3, \mathrm{dim} = C)$
// Step 4: Weighted Feature Fusion and Residual Addition
Broadcast multiply and add the residual input:
  $F_{out}^{SAFM} = F_{in}^{SAFM} + (\mathbf{w}_1 \odot Z_1) + (\mathbf{w}_2 \odot Z_2) + (\mathbf{w}_4 \odot Z_4)$
Return F_{out}^{SAFM}

3.3. Fission reorganization module

Moiré patterns, especially in high-resolution images, typically exhibit uneven distribution, with significant differences in texture density, frequency, and scale across different regions (Sun et al., 2018). Traditional image segmentation methods often struggle to simultaneously handle stripe features of varying densities. To address this challenge, we designed the Fission Reorganization Module (FRM), as shown in Fig. 4. Specifically, this module partitions the feature map into four sub-blocks. After being processed by a gating network, these blocks are routed into four distinct categories: flat regions for color cast calibration, dense high-frequency fine stripes, larger-scale dense stripes, and sparse low-frequency large ripples. Subsequently, they are directed into their corresponding specialized branches, where specific parameters are utilized for adaptive feature extraction. Finally, the original spatial dimensions are restored through reorganization.

3.3.1. Spatial fission and patch-level MoE routing

To enable the convolutional kernels to focus on moiré features in local regions and enhance the discriminative ability for different stripe densities, the FRM divides the feature map into smaller blocks and processes them block by block. As shown in Fig. 5, the feature map $F_{in}^{FRM} \in \mathbb{R}^{C\times H\times W}$ is evenly divided, and we define the splitting operator $S(\cdot)$, which maps F_{in}^{FRM} to a set of four overlapping sub-blocks $\mathcal{P}$:

$$\mathcal{P} = \{P_{i,j} \mid i,j \in \{1,2\}\} = S(\mathrm{Pad}(F_{in}^{FRM}, \delta)), \tag{5}$$

where δ represents the width of the overlapping pixels, and the size of each sub-block $P_{i,j}$ is $\left(\frac{H}{2}+2\delta\right)\times\left(\frac{W}{2}+2\delta\right)$. This overlap provides the necessary condition for mitigating boundary artifacts during the subsequent concatenation operation.

The resulting sub-blocks $\mathcal{P}$ are then fed into the Gating Network, where the most optimal computational path for each sub-block is selected from four specialized expert branches $\mathcal{E} = \{E_1, E_2, E_3, E_4\}$. To cater to varying moiré scales and frequencies, these experts are explicitly configured with distinct receptive fields, including 1×1 (for color cast calibration), 3×3 and 5×5 (for capturing dense, high-frequency fine stripes), and dilated 3×3 convolutions (for modeling sparse, low-frequency large ripples), as depicted in Fig. 4. For each sub-block $P_{i,j}$,

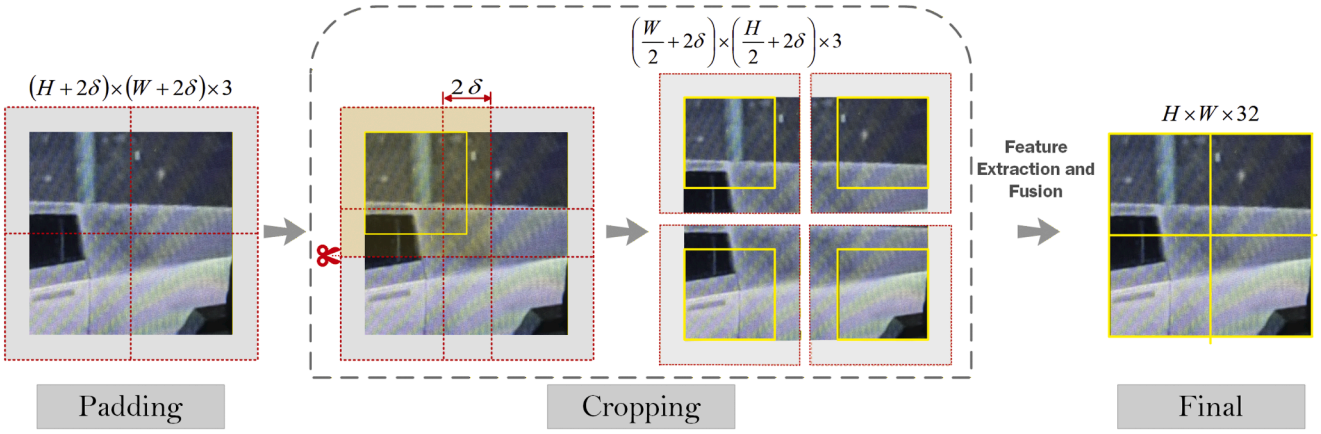

Fig. 5. Illustration of the spatial fission and reorganization strategy in FRM. The input feature map is first padded and divided into overlapping sub-blocks to preserve local context (Fission). After independent processing, the overlapping borders are cropped to eliminate boundary artifacts, and the valid regions are reorganized into the original spatial dimensions (Reorganization).

Algorithm 2 Patch-level MoE routing mechanism in FRM.

Input: Intermediate feature map $F_{in}^{FRM} \in \mathbb{R}^{C\times H\times W}$, Overlap size δ
Output: Reorganized feature map $F_{out}^{FRM} \in \mathbb{R}^{C_{out}\times H\times W}$

Initialize: Expert branches $\mathcal{E} = \{E_1, E_2, E_3, E_4\}$, Gating Network $\mathcal{G}$
// Step 1: Feature Fission (Partitioning with overlap)
Divide F_{in}^{FRM} into 4 overlapped blocks: $\mathcal{P} = \{P_{TL}, P_{TR}, P_{BL}, P_{BR}\}$
Initialize empty list O for processed blocks
// Step 2: Dynamic Routing and Expert Processing
for each block $P_{i,j}$ in $\mathcal{P}$ **do**
  Compute routing logits: $L = \mathcal{G}(P_{i,j})$ *// $L \in \mathbb{R}^4$*
  Determine target expert index via Hard Routing:
    $k = \arg\max(L)$
  Process block through the selected expert:
    $\hat{P}_{i,j} = E_k(P_{i,j})$
  Append $\hat{P}_{i,j}$ to O
end for
// Step 3: Feature Reorganization (Cropping and Stitching)
Crop overlap region δ from all blocks in O to get valid regions
Stitch the top row: $Top = \text{Concat}(\hat{P}_{TL}^{crop}, \hat{P}_{TR}^{crop}, \text{dim} = W)$
Stitch the bottom row: $Bottom = \text{Concat}(\hat{P}_{BL}^{crop}, \hat{P}_{BR}^{crop}, \text{dim} = W)$
Concatenate rows to form the full map:
  $F_{raw} = \text{Concat}(Top, Bottom, \text{dim} = H)$
Return Refined feature map $F_{out}^{FRM} = \mathcal{R}(F_{raw})$

the hard-routing transformation process is formulated as follows:

$$k = \arg\max\left(\mathcal{G}(P_{i,j})\right), \tag{6}$$

$$\hat{P}_{i,j} = E_k(P_{i,j}), \quad k \in \{1,2,3,4\}, \tag{7}$$

where $\mathcal{G}(\cdot)$ denotes the Gating Network module. To address the non-differentiability of the arg max function during backpropagation, we utilize the Straight-Through Estimator (STE) approximation (Bengio et al., 2013) to allow for end-to-end training. By utilizing these heterogeneous experts, the network achieves highly precise recognition of diverse moiré features in local regions. The detailed execution of this patch-level Mixture of Experts (MoE) is summarized in Algorithm 2.

3.3.2. Reorganization and refinement

Finally, we need to reassemble the processed local responses $\hat{P}_{i,j}$ into global features. To eliminate the overlapping redundancy introduced by the partitioning, we first apply the cropping operator Crop(·), and then use the concatenation operator Concat(·) to restore the original spatial dimensions:

$$F_{raw} = \begin{bmatrix} \text{Crop}(\hat{P}_{1,1}) & \text{Crop}(\hat{P}_{1,2}) \\ \text{Crop}(\hat{P}_{2,1}) & \text{Crop}(\hat{P}_{2,2}) \end{bmatrix}. \tag{8}$$

To smooth the concatenation boundaries and enhance cross-channel semantic fusion, we feed the reassembled features into the Bottleneck Block $\mathcal{R}(\cdot)$:

$$F_{out}^{FRM} = \mathcal{R}(F_{raw}). \tag{9}$$

In this module, $\mathcal{R}$ consists of a set of convolutional layers. The entire design of the FRM ensures that the model can adaptively address moiré frequencies through localized differential expert processing.

3.4. Loss function

In this section, to guide the model in precise pixel-level localization of the moiré regions, we use a combination of Binary Cross Entropy Loss (Long et al., 2015; Xie & Tu, 2017) and Dice Loss (Milletari et al., 2016) as the composite loss function (Yeung et al., 2022).

3.4.1. BCE loss

Since moiré detection is fundamentally a pixel-level binary classification task, we utilize BCE Loss as the basic supervision signal to guide the network in distinguishing moiré pixels from the background. It quantifies the pixel-wise discrepancy between the predicted probability map P and the ground truth mask Y, formulated as:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right], \tag{10}$$

where N is the total number of pixels, $y_i \in \{0,1\}$ denotes the true label of the ith pixel, and $p_i \in [0,1]$ represents the predicted probability output by the model.

3.4.2. Dice loss

To address the severe class imbalance between the moiré regions and the background, and to optimize the region-level overlap, we introduce Dice Loss. This effectively guides the model to align its segmentation boundaries closely with the real moiré patterns. The corresponding expression is written as:

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{N} p_i y_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} y_i + \epsilon}, \tag{11}$$

where ϵ is a smoothing factor used for numerical stability.

3.4.3. Total loss

To jointly leverage the advantages of pixel-wise classification and region-level structural optimization, the total loss of our proposed method is formulated as a weighted combination of BCE Loss and Dice Loss:

$$\mathcal{L}_{total} = \lambda \cdot \mathcal{L}_{BCE} + (1 - \lambda) \cdot \mathcal{L}_{Dice}, \tag{12}$$

where λ is a balancing weight coefficient. By dynamically balancing these two constraints, the network effectively enhances the structural coherence of the moiré patterns while ensuring pixel-level classification accuracy.

4. Experiment

4.1. Dataset

We conducted experiments on our proposed UHDMD dataset and the publicly available MoireScape dataset introduced by Yang et al. (2023).

4.1.1. UHDMD dataset

Motivation and Necessity: Although significant progress has been made in moiré pattern processing (Zheng et al., 2020), existing benchmark datasets primarily rely on synthetic data (Yuan et al., 2019) and generally suffer from low resolutions (Sun et al., 2018). These synthetic datasets often fail to accurately simulate the complex physical interactions between display sub-pixels and camera sensor arrays, nor can they capture the intricate features of moiré patterns on ultra-high-resolution screens. Typically generated by mathematical superposition, synthetic moiré patterns lack the nonlinear distortions, sensor noise, and varying lighting conditions inherent in real-world scenarios. To bridge this gap and facilitate the development of robust detection algorithms, we constructed the **U**ltra-**H**igh-**D**efinition **M**oiré **D**etection (UHDMD) dataset. To the best of our knowledge, UHDMD is the first real-world ultra-high-resolution dataset dedicated to moiré texture detection tasks.

Data Acquisition Pipeline: Prior research has established that the combination of imaging devices, display panels, and geometric factors (e.g., shooting angles and distances) significantly impacts the morphology of moiré patterns (Saveljev et al., 2018). Consequently, we employed multiple smartphones (e.g., OnePlus Ace 5, iPhone 15 Pro -Max, Xiaomi 16) to acquire data from various display devices (e.g., Lenovo Legion Y9000P, MacBook Air, Lenovo Legion Y7000P), as detailed in Table 1. To eliminate motion blur and accidental pixel shifts, all smartphones were mounted on dedicated stabilization stands, as shown in Fig. 6. We utilized voice-controlled shutter release and a wireless mouse to remotely switch display images, thereby avoiding mechanical vibrations during the capture process and ensuring precisely aligned image pairs. Furthermore, we binarized the captured pure moiré patterns to generate high-quality binary masks for training, as demonstrated in Fig. 7.

Detailed Characterization of UHDMD: The dataset comprises a total of 4625 image pairs, split into 4291 pairs for training and 334 pairs for testing. Each pair consists of a moiré-contaminated image and its corresponding pixel-aligned ground truth mask. Table 2 provides a comprehensive comparison between UHDMD and existing datasets. The key features of UHDMD are summarized as follows:

- **Ultra-high resolution**: UHDMD is the first non-synthetic 4K moiré detection dataset. Unlike previous works, it maintains high-frequency details essential for dense prediction tasks.
- **Content diversity**: The dataset encompasses a rich variety of scenes, each exhibiting moiré textures with distinct frequency distributions and spatial characteristics.
- **Real-world acquisition**: All images in UHDMD are captured directly from screens using smartphone cameras, faithfully simulating real-life image acquisition scenarios and preserving authentic optical distortions.

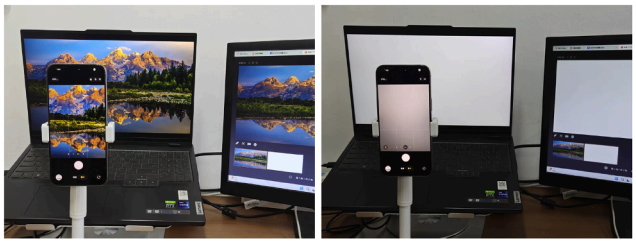

Fig. 6. Experimental setup for ultra-high-resolution moiré image acquisition.

Table 1
Capture devices and display specifications used for UHDMD data acquisition.

Mobile Phone	Shooting Resolution	Digital Screen	Display Resolution
OnePlus Ace 5	50 MP	Legion Y9000P	2560 × 1600
iPhone 15 Pro -Max	48 MP	MacBook Air	2880 × 1864
Xiaomi 16	50 MP	Legion Y7000P	2560 × 1600

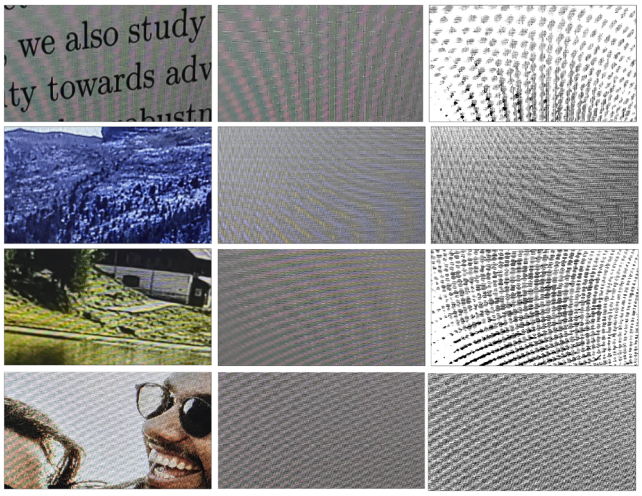

Fig. 7. Sample images from the UHDMD dataset: (Left) Captured moiré-contaminated image, (Middle) Isolated moiré pattern, and (Right) Corresponding binarized ground truth mask.

Table 2
Comparison between the MoiréScape and the proposed UHDMD datasets.

Dataset	Resolution	Images	UHD	Real	Diverse
MoiréScape	640 × 480	50,000	NO	NO	YES
UHDMD	3840 × 2160	4625	YES	YES	YES

4.1.2. MoireScape dataset

The MoiréScape dataset (Yang et al., 2023) contains only low-resolution synthetic moiré pattern images, with a total of 50,000 image pairs. In this study, to balance the evaluation scale, we randomly selected 18,000 image pairs as the training set and 8950 image pairs as the test set.

4.2. Evaluation metrics

To comprehensively evaluate the performance of SFRNet and other mainstream models, we employ widely adopted segmentation metrics,

including Intersection over Union (IoU) (Everingham et al., 2010), Precision, Recall, and F1-score (Manning et al., 2008). These metrics measure the model's capability to precisely locate and recognize moiré textures at the pixel level. The calculation formulas for these metrics are defined as follows:

$$IoU = \frac{TP}{TP + FP + FN}; \tag{13}$$

$$Precision = \frac{TP}{TP + FP}; \tag{14}$$

$$Recall = \frac{TP}{TP + FN}; \tag{15}$$

$$F1\text{-}score = \frac{2 \times Precision \times Recall}{Precision + Recall}, \tag{16}$$

where TP (True Positive) represents the number of moiré pixels correctly predicted by the model, FP (False Positive) denotes the number of background pixels incorrectly classified as moiré patterns, and FN (False Negative) is the number of actual moiré pixels missed by the model.

4.3. Implementation details

The proposed framework was implemented in PyTorch and trained on a server equipped with two NVIDIA RTX 3090 GPUs. We utilized the Adam optimizer with an initial learning rate of 2×10^{-4}. The training images were randomly cropped into patches of 512×512 pixels, with a batch size of 8. For the spatial fission process in the FRM, the overlap parameter δ is empirically set to 16 to ensure sufficient boundary context. A dynamic learning rate scheduler was employed: if the IoU score did not improve for five consecutive epochs, the learning rate was decayed by a factor of 0.5. The model was trained for a total of 100 epochs. Furthermore, the balancing weight λ in the total loss function is empirically set to 0.3 across experiments.

4.4. Quantitative and qualitative comparison

Since moiré pattern detection is fundamentally a pixel-level binary classification task, we conducted a comprehensive comparison of SFRNet against several representative state-of-the-art (SOTA) semantic segmentation algorithms, including classic architectures such as DeepLabV3+ (Chen et al., 2018), SegNet (Badrinarayanan et al., 2017), and HRNet (Wang et al., 2021), as well as recently proposed cutting-edge models like InSPyReNet (Kim et al., 2022) and BiRefNet (Zheng et al., 2024). It is worth noting that we also considered recent joint-learning methods like MoireDet (Yang et al., 2023). However, due to their strict requirement for perfectly paired clean images and their prohibitive memory consumption at UHD resolutions, they cannot be directly integrated into our binary detection evaluation benchmark. To ensure a fair comparison, all baseline models were retrained under identical experimental settings.

4.4.1. Quantitative evaluation

To comprehensively evaluate the detection performance of our proposed model, we conducted quantitative experiments on both real-world and synthetic datasets. The comparative results are summarized in Table 3.

Results on UHDMD Dataset: As shown in the table, SFRNet demonstrates superior performance on the real-world UHD dataset. Specifically, SFRNet achieves an IoU of **0.6190** and an F1-score of **0.8845**, significantly outperforming the runner-up SegNet (IoU 0.6054) and HRNet (IoU 0.6046). Notably, while HRNet exhibits the highest recall rate (0.9978), its precision remains the lowest (0.6154). This discrepancy indicates that such general-purpose models suffer from severe over-segmentation, tending to misclassify complex background textures as moiré patterns. Furthermore, recent advanced models like InSPyReNet and BiRefNet exhibit suboptimal performance on this specific task (yielding IoU scores of 0.5450 and 0.4862, respectively). This is primarily because these models are optimized for general object semantics, struggling to localize the high-frequency and spatially non-uniform artifacts inherent in 4K images. In contrast, SFRNet achieves the highest Precision (**0.6286**) and IoU while maintaining a comparable ultra-high Recall (0.9942). This balance verifies the effectiveness of the FRM module in handling spatial non-uniformity, ensuring accurate boundary localization. Therefore, although the numerical increase in IoU is relatively small, the significant improvement in Precision directly reduces false positives. In real-world tasks like anti-spoofing, avoiding false alarms is crucial because systems cannot afford to wrongly identify real images as fake ones. By effectively minimizing these errors, our method proves its practical reliability for actual deployment.

Results on MoiréScape Dataset: Despite the relatively simple texture distribution of the synthetic MoiréScape dataset, SFRNet maintains a substantial advantage, achieving an IoU of 0.9338 and an F1-score of 0.9768. However, recent advanced models such as InSPyReNet and BiRefNet suffer a severe performance drop, yielding IoU scores of only 0.6246 and 0.6397, respectively. This phenomenon indicates that these models rely on recognizing common object shapes and struggle to detect moiré patterns in images. Compared to traditional baselines like DeepLabV3+ (IoU 0.8527) and HRNet (IoU 0.8526), our method brings an IoU improvement of over **8.1%**. This further confirms that SFRNet not only relies on object semantics but is truly adept at capturing multi-scale frequency features, consistently outperforming traditional algorithms and the latest advanced methods across different data domains.

4.4.2. Qualitative evaluation

To qualitatively validate the superiority of SFRNet in visual perception, we performed a visual comparison between SFRNet and the baseline models on the UHDMD dataset. As shown in Fig. 8, existing methods have limited generalization capability in challenging 4K scenes: recently advanced models like BiRefNet and InSPyReNet struggle to capture the delicate, high-frequency repetitive structures, producing highly distorted or fragmented masks that lose the topological integrity of the ripples. Meanwhile, DeepLabV3+ and SegNet tend to generate over-smoothed masks, and HRNet produces severe false positives due to overfitting to complex background textures. In contrast, SFRNet achieves outstanding scale robustness and edge fidelity. Thanks to the aggregation of multi-scale features by SAFM and the handling of local feature heterogeneity by FRM, our model can accurately distinguish moiré artifacts from complex background textures, preserving the clear topological boundaries of both large-scale color distortions and fine high-frequency ripples.

4.5. Ablation study

We conducted ablation experiments on the UHDMD dataset to verify the effectiveness of the proposed modules and structural hyperparameters. The baseline model is constructed by removing SAFM and FRM from SFRNet.

4.5.1. Effectiveness of components

To evaluate the individual contributions of SAFM and FRM to the overall detection performance, we incrementally integrated them into the plain baseline network. The quantitative results of this ablation study are summarized in Table 4. As observed, the baseline model achieves an IoU of 60.29%. Integrating FRM or SAFM individually only results in slight improvements, bringing the IoU to 60.55% and 60.67%, respectively. This indicates that neither global scale awareness nor local feature refinement alone is sufficient to significantly enhance the model's ability to handle the complex task of ultra-high-definition moiré detection. However, when both modules are combined, a significant performance improvement is achieved, with the IoU reaching 61.90% and the F1-score reaching 88.45%. This demonstrates that SAFM and FRM are complementary: SAFM provides moiré features at different scales,

Table 3
Quantitative comparison of different methods on the **UHDMD** (Real-world) and **MoiréScape** (Synthetic) datasets. The best results are highlighted in **bold**.

Method	UHDMD Dataset (Real-world)				MoiréScape Dataset (Synthetic)			
	IoU	F1-score	Recall	Precision	IoU	F1-score	Recall	Precision
BiRefNet (Zheng et al., 2024)	0.4862	0.6432	0.7373	0.6783	0.6397	0.7069	0.6759	0.9261
InSPyReNet (Kim et al., 2022)	0.5450	0.7796	0.8463	0.6159	0.6246	0.7276	0.7136	0.8409
DeepLabV3+ (Chen et al., 2018)	0.6013	0.8641	0.9795	0.6189	0.8527	0.9656	0.9997	0.8528
HRNet (Wang et al., 2021)	0.6046	0.8643	**0.9978**	0.6154	0.8526	0.9658	**1.0000**	0.8526
SegNet (Badrinarayanan et al., 2017)	0.6054	0.8736	0.9856	0.6170	0.9233	0.9718	0.9799	0.9410
SFRNet (Ours)	**0.6190**	**0.8845**	0.9942	**0.6286**	**0.9338**	**0.9768**	0.9845	**0.9477**

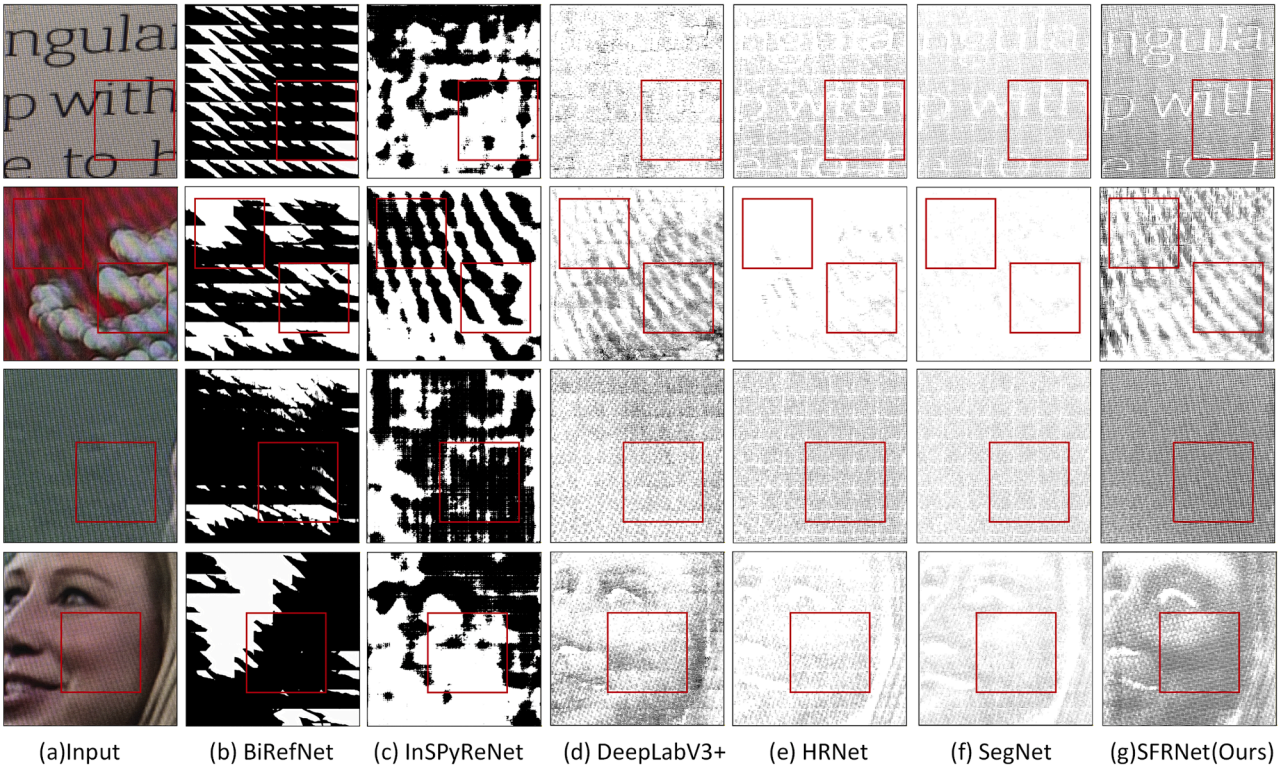

Fig. 8. Visual comparisons on the UHDMD dataset. From left to right: (a) Input Moiré images, (b) BiRefNet, (c) InSPyReNet, (d) DeepLabV3+, (e) HRNet, (f) SegNet, and (g) SFRNet (Ours).

Table 4
Ablation study of different components on the UHDMD dataset.

SAFM	FRM	IoU (%)	F1-score (%)	Recall (%)	Precision (%)
×	×	60.29	86.23	96.19	61.80
×	✓	60.55	86.62	96.83	61.74
✓	×	60.67	86.52	96.55	61.96
✓	✓	**61.90**	**88.45**	**99.42**	**62.86**

Table 5
Impact of different cropping strategies on detection performance.

Strategy	IoU (%)	F1-score (%)	Recall (%)	Precision (%)
Center Crop	60.93	87.36	98.02	61.64
Random Crop	**61.90**	**88.45**	**99.42**	**62.86**

while FRM refines local details. Together, they enable the network to effectively handle multi-scale frequency aliasing.

4.5.2. *Impact of cropping strategies*

To investigate the influence of different spatial augmentation strategies on the model's spatial generalization, we compared the center cropping method with our default random cropping strategy. The qualitative visual comparisons and quantitative metrics are presented in Fig. 9 and Table 5, respectively. As illustrated in Fig. 9, the model trained with center cropping exhibits a strong center bias, making it prone to missing moiré artifacts located in the peripheral regions of the image. In contrast, random cropping endows the network with translation invariance. This qualitative analysis is further substantiated by the quantitative results: compared to center cropping (IoU 60.93%), random cropping leads to an improvement of IoU by 0.97% and effectively boosts the recall to 99.42%. This confirms that ensuring spatial diversity during training is crucial for robust generalization.

4.5.3. *Structural hyperparameter analysis*

To justify our network design, we further conducted ablation studies on the structural hyperparameters of the proposed modules, specifically the number of multi-scale branches in SAFM, the number of expert branches in FRM, and the overlap parameter δ.

The quantitative results for the structural branches are detailed in Table 6.

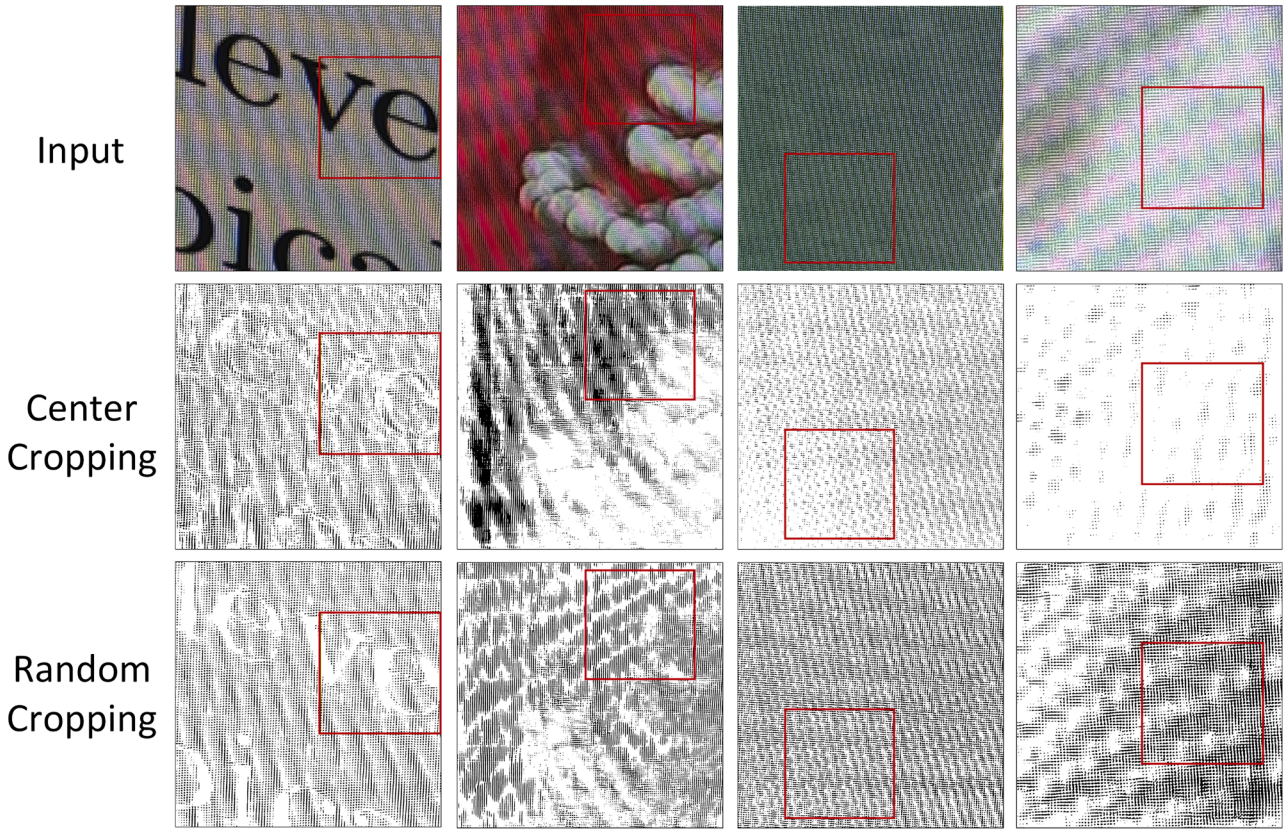

Fig. 9. Qualitative comparison of different cropping strategies. As highlighted by the red boxes, the model trained with center cropping often fails to detect moiré patterns in peripheral regions due to center bias. In contrast, the random cropping strategy successfully preserves detection capabilities across the entire image spatial domain.

Number of Scale Branches in SAFM: We evaluated the impact of reducing the SAFM branches from three ($1\times, 2\times, 4\times$) to two ($1\times, 2\times$). As shown in Table 6, the dual-branch configuration leads to a performance drop, with the IoU decreasing to 60.81%. This confirms that the $4\times$ downsampling branch is indispensable for capturing large-scale color ripples, and the three-branch pyramid achieves the optimal balance for scale-aware fusion.

Number of Experts in FRM: We also investigated the effect of the expert capacity in the patch-level MoE routing mechanism. We compared our default 4-expert setting with 1-expert (equivalent to a standard convolutional branch without routing) and 2-expert configurations. The results indicate that a single expert (IoU 60.48%) fails to address the heterogeneous moiré frequencies. Increasing the capacity to 2 experts provides a noticeable improvement (IoU 61.04%), but the 4-expert configuration yields the best performance (IoU 61.90%). This demonstrates that a sufficient number of specialized experts is required to adequately decouple complex moiré patterns from diverse background textures. Furthermore, we evaluated an 8-expert configuration. However, the results show that increasing the number of experts to 8 leads to a performance drop (IoU decreases to 61.51%). Having too many experts causes routing redundancy and overfitting to background noise. It also significantly increases the computational cost. Therefore, the 4-expert setting is chosen as the best balance between detection accuracy and computational efficiency.

Sensitivity of Overlap Parameter δ: We also analyzed the selection and sensitivity of the overlap parameter δ used in FRM cropping. Theoretically, since our expert branches use 5×5 and dilated 3×3 convolutions (equivalent to a 5×5 receptive field), an overlap is needed to preserve local context and avoid boundary artifacts during patch processing. Our experiments on the UHDMD dataset support this: setting $\delta = 0$ (no overlap) causes a slight performance drop (IoU 61.78%) due to boundary stitching errors. However, when we increase δ beyond the minimum requirement (e.g., $\delta = 8, 16, 24, 32$), the detection performance remains stable (IoU ranges from 61.81% to 61.90%). This confirms that our network is robust and not sensitive to the exact value of δ. We set $\delta = 16$ as the default configuration to maintain boundary quality without adding unnecessary computation.

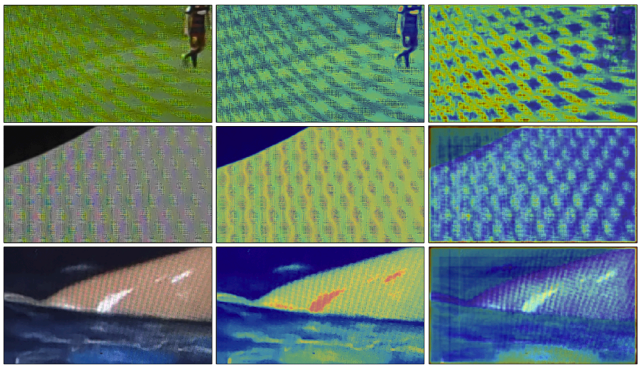

Fig. 10. Feature visualization comparison under diverse real-world scenarios. (Left column) Original input moiré images. (Middle column) Feature activations without SAFM and FRM, showing severe background interference. (Right column) Refined feature activations with SAFM and FRM, demonstrating precise decoupling of moiré patterns from complex UHD backgrounds.

Table 6
Ablation study on the structural hyperparameters of SAFM (number of scale branches) and FRM (number of expert branches).

Module Settings	Variants	IoU (%)	F1-score (%)	Recall (%)	Precision (%)
SAFM Branches	2 Branches	60.81	86.78	96.94	61.97
	3 Branches (Ours)	**61.90**	**88.45**	**99.42**	**62.86**
FRM Experts	1 Expert	60.48	86.20	96.05	61.92
	2 Experts	61.04	87.45	98.12	61.73
	4 Experts (Ours)	**61.90**	**88.45**	**99.42**	**62.86**
	8 Experts	61.51	88.14	99.14	61.82

Table 7
Computational efficiency comparison. Latency and memory are measured during the inference of a full 3840×2160 image using the sliding-window strategy on an RTX 3090 GPU.

Method	Params (M) ↓	FLOPs (G) ↓	Latency (s) ↓	Memory (GB) ↓
DeepLabV3+	59.23	177.07	**0.63**	0.52
SegNet	29.44	321.05	0.97	0.77
HRNet	28.54	**82.38**	1.53	**0.32**
InSPyReNet	90.52	378.80	2.94	1.20
BiRefNet	214.96	604.77	4.48	1.57
SFRNet (Ours)	**7.78**	372.02	3.44	0.92

4.5.4. Feature visualization analysis

To intuitively demonstrate the internal mechanisms of our proposed modules across diverse scenarios, we visualize the intermediate feature maps of SFRNet before and after the application of SAFM and FRM. Fig. 10 presents several representative cases, including challenging scenes with varying moiré frequencies and complex background interference.

Without the enhancement of our core modules (middle column), the network's feature activations are diffuse. The network struggles to disentangle the moiré artifacts from the underlying high-frequency content, leading to severe background false activations (widespread red/yellow regions). However, after being processed by the cross-scale dynamic fusion of SAFM and the patch-level MoE routing of FRM (right column), the network's attention undergoes a dramatic and precise shift. Across all showcased scenarios, the high activations (red/yellow regions) strictly align with the structural layout of the moiré fringes, while responses in the background areas are significantly suppressed (dark blue). This striking visual contrast provides compelling evidence that our dual-module design effectively decouples complex moiré frequencies from background details, intrinsically explaining the quantitative improvements observed in the ablation studies.

4.5.5. Computational efficiency analysis

To evaluate the computational efficiency of our method, we compared SFRNet with baseline models. The evaluation metrics include parameters (Params), FLOPs (measured at a 512×512 input size), latency (average processing time for a single 4K image), and peak GPU memory usage during sliding-window inference. As shown in Table 7, SFRNet has the lowest parameter count (7.78 M) among all the evaluated models. Due to the patch-level dynamic routing and multi-scale feature fusion, its FLOPs and latency are higher than those of traditional lightweight networks (e.g., HRNet). However, compared to recent models like BiRefNet, SFRNet has a much lower computational cost. Furthermore, the sliding-window strategy limits the peak memory usage to 0.92 GB. This shows that SFRNet achieves a good balance between detection accuracy and computational cost for UHD scenarios.

5. Conclusion

In this paper, we have presented a comprehensive study on the challenging task of ultra-high-definition (UHD) moiré pattern detection. To address the scarcity of high-quality real-world benchmarks, we constructed **UHDMD**, the first non-synthetic UHD dataset dedicated to this task. Based on this benchmark, we proposed **SFRNet**, a novel architecture that integrates a Scale-Aware Fusion Module (SAFM) and a Fission Reorganization Module (FRM). By combining cross-scale dynamic fusion with a patch-level Mixture of Experts (MoE) routing mechanism, our network effectively captures multi-scale moiré features and handles severe spatial non-uniformity in 4K images. Extensive experiments demonstrate that SFRNet significantly outperforms state-of-the-art segmentation methods, establishing a new robust baseline for real-world UHD moiré detection. Theoretically, our approach provides a new perspective for separating complex high-frequency artifacts from background details. Practically, SFRNet effectively reduces false detections and achieves high precision on 4K images, making it highly reliable for real-world applications like digital forensics and biometric anti-spoofing.

Limitations and Future Work: Despite the promising performance, there remains room for improvement. First, the high computational complexity required to process massive numbers of pixels in UHD images poses a challenge for real-time deployment on resource-constrained edge devices. Second, the construction of the UHDMD dataset currently relies on labor-intensive manual capture and pixel-level annotation, which limits rapid scalability. In future work, we aim to explore lightweight network designs to optimally balance efficiency and accuracy. Furthermore, we will investigate more cost-effective strategies to facilitate the rapid expansion of the UHDMD dataset, thereby continuously advancing the research of real-world moiré pattern detection.

CRediT authorship contribution statement

Pengyue Xu: Conceptualization, Methodology, Software, Data curation, Formal analysis, Writing – original draft, Writing – review & editing; **Shuang Chen:** Conceptualization, Data curation, Validation, Formal analysis, Writing – review & editing; **Qixian Zhang:** Methodology, Validation, Formal analysis, Investigation, Writing – review & editing; **Ning Xie:** Validation, Investigation, Writing – review & editing; **Hongbin Wang:** Conceptualization, Resources, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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